

ISTE240-Group9 Project Documentation



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1 Introduction

The objective of this particular project is the development of the Field Booking System, a web-based application intended for the effective management of the booking season for sports fields. Such a system would come in handy for organizations such as school districts, municipalities, sports clubs, and individuals.

The main aim of this project is to ensure that a scalable and user-friendly web application is developed, enabling users to view field availability, reserve fields, and manage bookings in a web-based environment. Users would be able to manage fields, while administrators would approve or reject bookings. The backend would be implemented using Java Spring.

2 Project Scope and Timeline

Scope

The Field Booking System will include:

1. User authentication and role-based access
2. Creating, modifying, and canceling bookings
3. Administrative management of fields and reservations

Expected Timeline

- Weeks 1–2: Requirement analysis and system design
- Weeks 3–5: Backend development (Spring Boot APIs, database setup)
- Weeks 6–7: Frontend development and UI integration
- Week 8: Testing, bug fixes, and final documentation

3 Key Features

User Registration and Login: Secure authentication for users and administrators.

Field Listings: Displaying Available Fields with Descriptions and Scheduled Times

Booking Management: User can add, modify and delete booking

Schedule Display Page: Availability Calendar - Visualizing Availability

Admin Dashboard: Tools for Administering Fields, Approval for Bookings, Running Conflicts

4 Implementation Details

Backend: Java Spring Boot

Frontend: To be determined

Database: MariaDB as Relational database

Tools:

- GitHub for version control
- Maven for dependency management
- IntelliJ IDEA as the IDE

5 Anticipated Challenges

Learning Curve with Spring Framework

Time Constraints